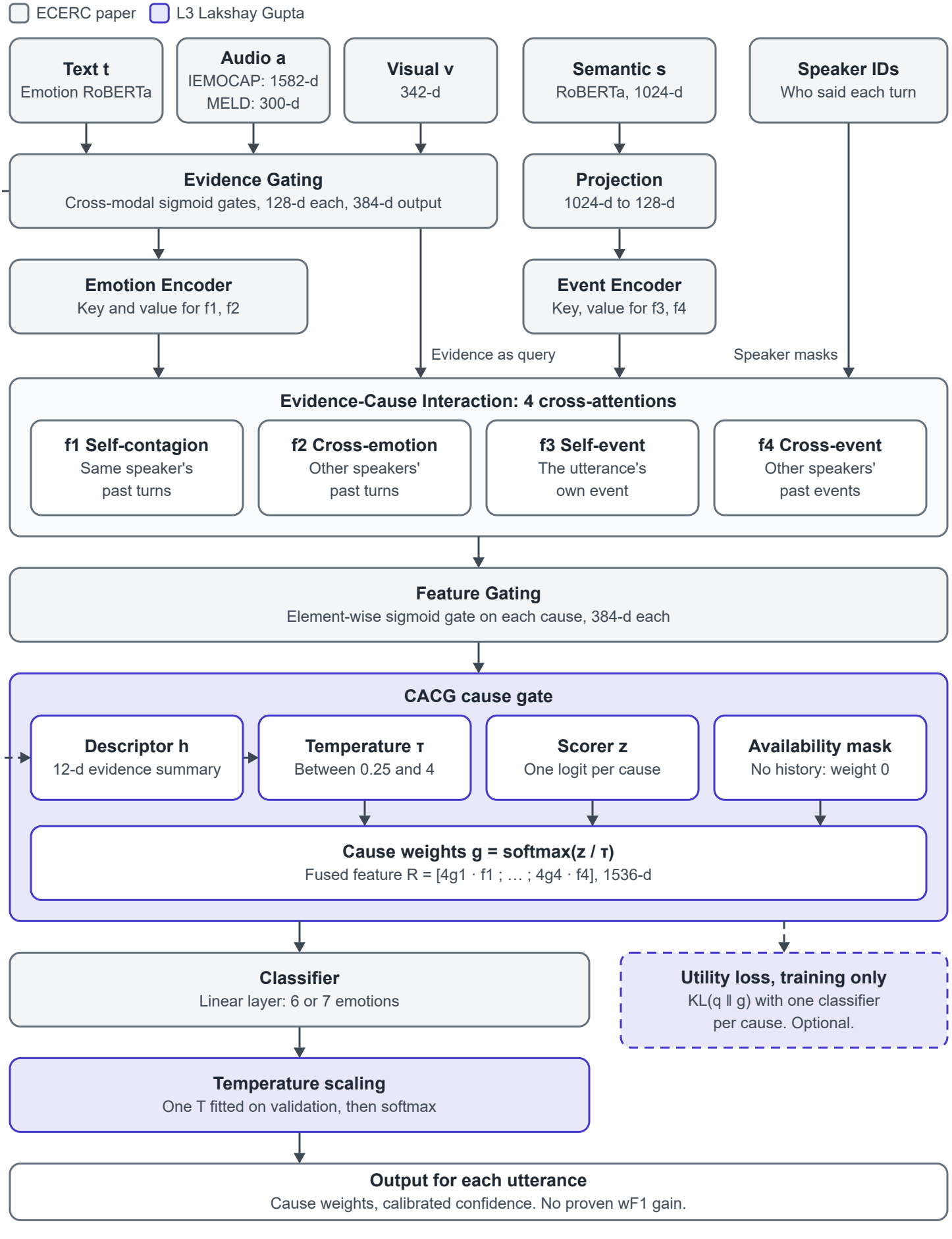

Limitation 3 alone: ECERC with the CACG cause gate
ECERC paper
L3 Lakshay Gupta
Text t
Emotion RoBERTa
Audio a
IEMOCAP: 1582-d
MELD: 300-d
Visual v
342-d
Semantic s
RoBERTa, 1024-d
Speaker IDs
Who said each turn
Evidence Gating
Cross-modal sigmoid gates, 128-d each, 384-d output
Projection
1024-d to 128-d
Emotion Encoder
Key and value for f1, f2
Event Encoder
Key, value for f3, f4
Evidence as query
Speaker masks
Evidence-Cause Interaction: 4 cross-attentions
f1 Self-contagion
Same speaker's past turns
f2 Cross-emotion
Other speakers' past turns
f3 Self-event
The utterance's own event
f4 Cross-event
Other speakers' past events
Feature Gating
Element-wise sigmoid gate on each cause, 384-d each
CACG cause gate
Descriptor h
12-d evidence summary
Temperature τ
Between 0.25 and 4
Scorer z
One logit per cause
Availability mask
No history: weight 0
Cause weights g = softmax(z / τ)
Fused feature R = [4g1 · f1 ; … ; 4g4 · f4], 1536-d
Classifier
Linear layer: 6 or 7 emotions
Utility loss, training only
KL(q ‖ g) with one classifier per cause. Optional.
Temperature scaling
One T fitted on validation, then softmax
Output for each utterance
Cause weights, calibrated confidence. No proven wF1 gain.